

Fields of Astrophysics

A Mathematical Journey Through the Cosmos

From the Big Bang to Black Holes, from Stellar Nurseries to Gravitational Waves —
every phenomenon is written in the language of mathematics.

May 16, 2026

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1. Introduction: Why Mathematics?

Astrophysics is, above all else, the audacious attempt to understand the universe using the same physics we discover in laboratories on Earth. What makes it remarkable is not just the scale — we are talking about distances of billions of light-years and timescales of billions of years — but the fact that *it works*. The same equations that describe a falling apple also govern the precession of Mercury’s orbit. The same quantum mechanics that powers our transistors explains why stars shine.

Curiosity. The observable universe contains roughly 2×10^{23} stars. That is more stars than grains of sand on all of Earth’s beaches combined — by a factor of about ten thousand.

This document tours the major sub-fields of astrophysics, presenting the key equations in each domain alongside plots and diagrams to build intuition. Strap in.

2. Cosmology: The Universe at Largest Scales

2.1 The Expanding Universe

In 1929, Edwin Hubble measured the recession velocities of galaxies and found a stunning linear relationship between distance d and velocity v :

$$v = H_0 d \quad (1)$$

where $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is the **Hubble constant**. This is not galaxies moving *through* space — it is space itself that is expanding.

2.2 The Friedmann Equations

The dynamics of the expanding universe are governed by Einstein’s General Relativity applied to a homogeneous, isotropic spacetime (the **FLRW metric**):

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \quad (2)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3} \quad (3)$$

Here $a(t)$ is the **scale factor**, ρ is the energy density, p is pressure, $k \in \{-1, 0, +1\}$ is the spatial curvature, and Λ is the **cosmological constant** (dark energy). The Hubble parameter is simply $H \equiv \dot{a}/a$.

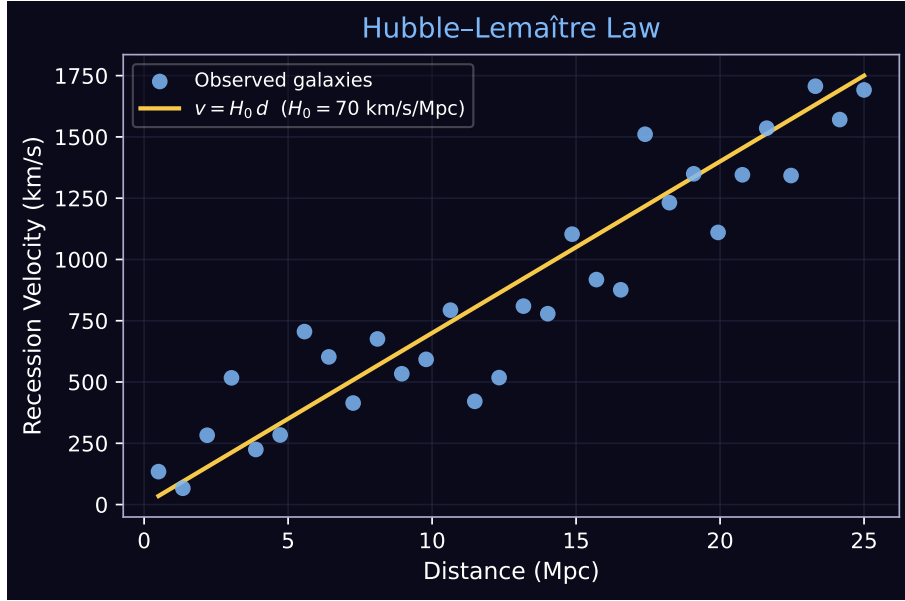


Figure 1: Hubble–Lemaître Law: recession velocity vs. distance for a simulated galaxy sample. The slope is $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Curiosity. Observations show that $\Omega_\Lambda \approx 0.68$, meaning roughly 68% of the universe’s energy budget is dark energy — a form of energy we have no microscopic understanding of whatsoever.

2.3 The Cosmic Microwave Background

The CMB is the thermal afterglow of the Big Bang, emitted $\sim 380,000$ years after the birth of the universe. Its angular temperature fluctuations $\delta T/T \sim 10^{-5}$ encode the initial conditions for all structure. We expand them in spherical harmonics:

$$\frac{\delta T}{T}(\hat{n}) = \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n}) \quad (4)$$

The **angular power spectrum** $C_\ell = \langle |a_{\ell m}|^2 \rangle$ peaks at multipoles corresponding to the *acoustic oscillations* of the primordial plasma:

3. Stellar Astrophysics

3.1 Hydrostatic Equilibrium

A star is a self-gravitating ball of plasma held up by its own pressure. The condition of **hydrostatic equilibrium** requires:

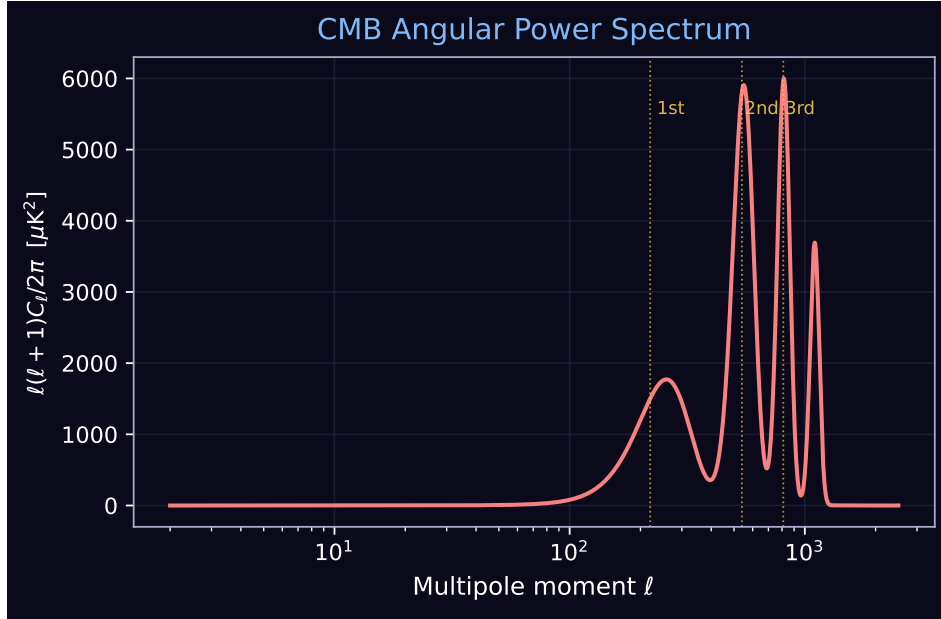


Figure 2: Approximate CMB TT power spectrum showing the characteristic acoustic peaks. The first peak at $\ell \approx 220$ corresponds to the angular scale of the sound horizon at recombination ($\sim 1^\circ$).

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2} \quad (5)$$

Combined with the equation of mass continuity $dM/dr = 4\pi r^2 \rho$ and an equation of state $P(\rho, T)$, these form the foundation of stellar structure theory.

3.2 The Virial Theorem

For a gravitationally bound system in equilibrium, the **Virial Theorem** states:

$$\langle E_{\text{kin}} \rangle = -\frac{1}{2} \langle E_{\text{grav}} \rangle \quad (6)$$

This implies that as a star loses energy (radiates), it actually *heats up* — a counterintuitive result with profound consequences for stellar evolution.

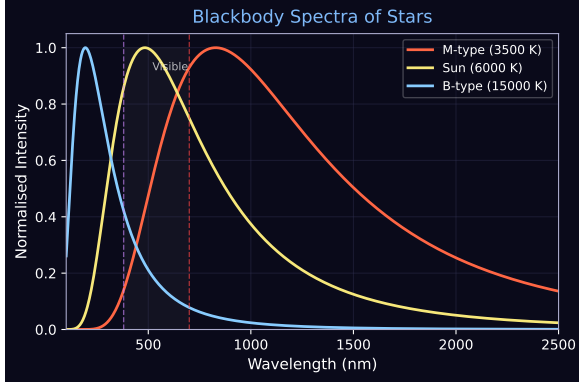
3.3 Blackbody Radiation and the HR Diagram

Stars radiate approximately as **blackbodies**. The spectral radiance is given by the **Planck function**:

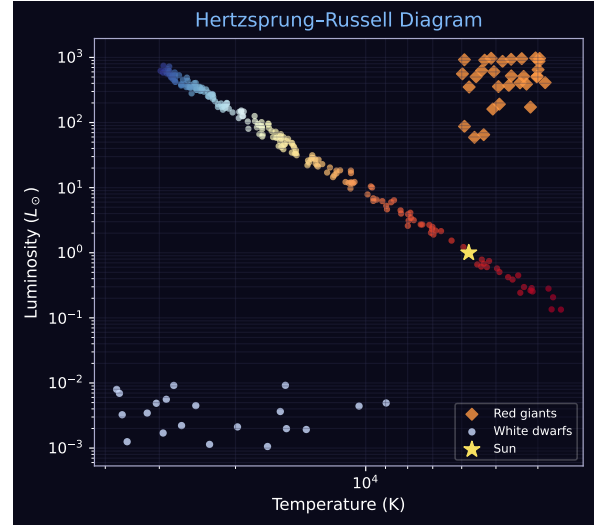
$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} \quad (7)$$

The total luminosity then follows from the **Stefan–Boltzmann law**:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \quad (8)$$



(a) Planck spectra for three stellar types. The peak wavelength shifts with temperature via Wien's law: $\lambda_{\text{max}} = b/T$, $b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$.



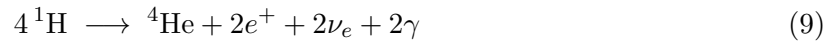
(b) Hertzsprung–Russell diagram. The *main sequence* is where hydrogen burns. Stars spend $\sim 90\%$ of their lives here before evolving to giants or white dwarfs.

Figure 3: Stellar radiation and classification.

Curiosity. The Sun will spend about 10 billion years on the main sequence. It is currently about halfway through. When it leaves, it will expand into a red giant large enough to engulf Earth's current orbit.

3.4 Nuclear Burning: The PP Chain

The Sun's energy comes from fusing hydrogen into helium via the **proton–proton (pp) chain**:



The energy released per reaction is $Q = \Delta m c^2 \approx 26.7 \text{ MeV}$. The Sun converts $\sim 4 \times 10^9 \text{ kg}$ of mass into energy every second.

4. General Relativity and Black Holes

4.1 The Einstein Field Equations

General Relativity describes gravity as the curvature of spacetime. The Einstein field equations read:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (10)$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor (encoding curvature), and $T_{\mu\nu}$ is the stress-energy tensor (encoding matter and energy). This single, elegant equation says: *spacetime curvature is sourced by energy-momentum*.

4.2 The Schwarzschild Metric

The simplest black hole solution (spherically symmetric, non-rotating) is the **Schwarzschild metric**:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad (11)$$

The **Schwarzschild radius** $r_s = 2GM/c^2$ marks the event horizon. For the Sun, $r_s \approx 3$ km.

4.3 Orbital Dynamics and the ISCO

The effective potential for a test particle in Schwarzschild spacetime is:

$$V_{\text{eff}}(r) = \left(1 - \frac{r_s}{r}\right) \left(1 + \frac{L^2}{r^2 c^2}\right) \quad (12)$$

where L is the specific angular momentum. Unlike Newtonian gravity, there is an **Innermost Stable Circular Orbit (ISCO)** at $r_{\text{ISCO}} = 3r_s$. Inside this radius, no stable orbit exists — matter must plunge in.

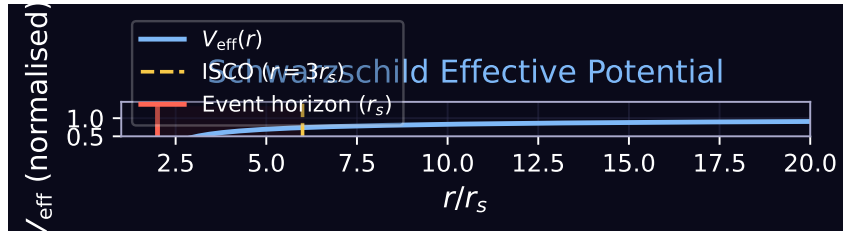


Figure 4: Effective potential in Schwarzschild spacetime. Inside the ISCO (yellow dashed line), circular orbits become unstable. Inside the event horizon (red solid line), *nothing* — not even light — can escape.

Curiosity. If you fell into a stellar-mass black hole, tidal forces would *spaghettify* you before you crossed the horizon. But if you fell into a supermassive black hole ($M \sim 10^9 M_\odot$), you would cross the horizon without noticing anything unusual — the tidal forces there can be weaker than Earth’s gravity.

5. Gravitational Wave Astronomy

5.1 The Quadrupole Formula

Accelerating masses radiate **gravitational waves**. The emitted power (in the weak-field, slow-motion approximation) is given by the **quadrupole formula**:

$$P = -\frac{G}{5c^5} \langle \ddot{Q}_{ij} \ddot{Q}^{ij} \rangle \quad (13)$$

where $Q_{ij} = \int \rho (x_i x_j - \frac{1}{3} \delta_{ij} r^2) dV$ is the mass quadrupole moment.

5.2 Strain and Detection

The observable quantity is the dimensionless **strain**:

$$h \equiv \frac{\Delta L}{L} = \frac{4G}{rc^4} \ddot{Q}_{ij} \quad (14)$$

For GW150914 (the first detection by LIGO in 2015), the strain was $h \sim 10^{-21}$ at Earth — meaning a change in arm length of $\sim 10^{-18}$ m, one-thousandth the diameter of a proton.

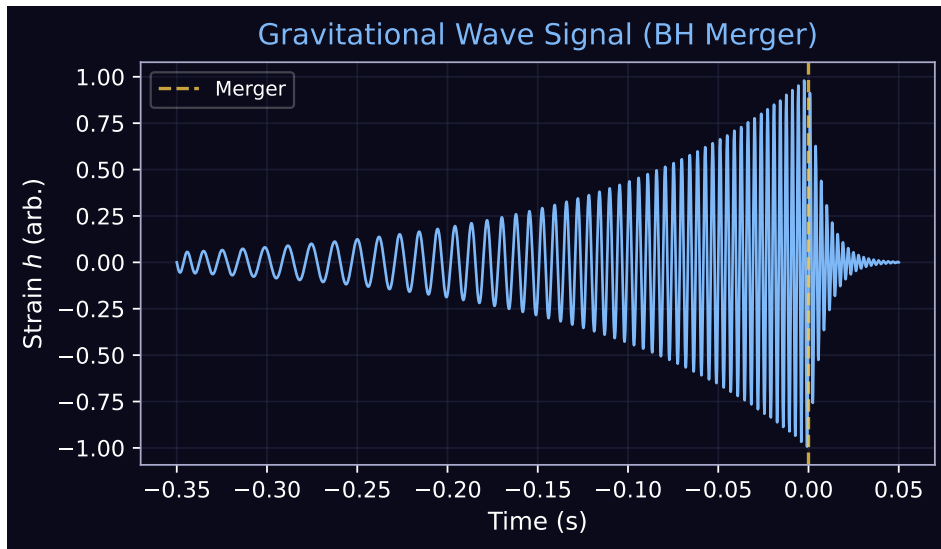


Figure 5: Simulated gravitational wave strain from a binary black hole merger. The amplitude and frequency both increase (*chirp*) as the black holes spiral inward, peaking at merger, then decaying as the remnant rings down.

Curiosity. The two merging black holes in GW150914 had a combined mass of $\sim 65 M_{\odot}$, but the final black hole was only $\sim 62 M_{\odot}$. Three solar masses — equivalent to 6×10^{30} kg — were radiated as gravitational waves in about *0.2 seconds*.

6. High-Energy Astrophysics

6.1 Synchrotron Radiation

Relativistic electrons spiralling in magnetic fields emit **synchrotron radiation**. The characteristic frequency is:

$$\nu_c = \frac{3}{4\pi} \frac{eB}{m_e c} \gamma^2 \sin \alpha \quad (15)$$

where $\gamma = E/m_e c^2$ is the Lorentz factor. This mechanism powers the jets of active galactic nuclei (AGN) and gamma-ray bursts.

6.2 The Eddington Luminosity

There is a maximum luminosity for a gravitationally bound, radiation-pressure-dominated object — the **Eddington Luminosity**:

$$L_{\text{Edd}} = \frac{4\pi G M m_p c}{\sigma_T} \approx 1.26 \times 10^{31} \left(\frac{M}{M_\odot} \right) \text{ W} \quad (16)$$

where σ_T is the Thomson cross-section. Exceeding L_{Edd} would blow the accreting material away, setting a limit on how quickly black holes can grow.

6.3 X-ray Binaries and Accretion

Accreting matter releases gravitational potential energy. The accretion luminosity is:

$$L_{\text{acc}} = \eta \dot{M} c^2 \quad (17)$$

where η is the radiative efficiency ($\eta \approx 0.1$ for standard thin-disk accretion, compared to $\eta \approx 0.007$ for nuclear burning). Accretion onto compact objects is the most efficient energy conversion process known in nature.

7. Galactic Astrophysics

7.1 The Milky Way and Dark Matter

Stars and gas in a galaxy orbit under the combined gravitational potential. Newton's law gives the circular velocity:

$$v_{\text{circ}}(r) = \sqrt{\frac{GM(r)}{r}} \quad (18)$$

If all mass were in the luminous disk, v_{circ} should fall as $r^{-1/2}$ beyond the disk. Instead, observations show flat rotation curves ($v_{\text{circ}} \approx \text{const}$), implying $M(r) \propto r$ — a vast halo of unseen **dark matter**.

Curiosity. The Milky Way's dark matter halo extends ~ 100 – 200 kpc, dwarfing the visible disk (~ 15 kpc). Dark matter makes up $\sim 27\%$ of the universe's total energy budget, yet we have never directly detected a dark matter particle.

7.2 The NFW Profile

The dark matter density profile of halos in simulations follows the **Navarro–Frenk–White (NFW) profile**:

$$\rho(r) = \frac{\rho_0}{\frac{r}{r_s} \left(1 + \frac{r}{r_s}\right)^2} \quad (19)$$

where r_s is the scale radius and ρ_0 is a characteristic density.

8. Exoplanet Science

8.1 Radial Velocity

The **Doppler wobble** of a star due to an orbiting planet produces a velocity semi-amplitude:

$$K_\star = \frac{m_p \sin i}{(M_\star + m_p)^{1/3}} \left(\frac{2\pi G}{P} \right)^{1/3} \frac{1}{\sqrt{1 - e^2}} \quad (20)$$

where P is the orbital period, e is the eccentricity, and i is the orbital inclination.

8.2 Transits and the Transit Depth

When a planet transits its host star, the fractional dimming is:

$$\delta = \left(\frac{R_p}{R_\star} \right)^2 \quad (21)$$

For an Earth-sized planet around a Sun-like star, $\delta \approx 84$ ppm — a precision only achievable from space (as Kepler and TESS demonstrated).

Curiosity. The Kepler Space Telescope discovered over 2,600 confirmed exoplanets by staring at a single patch of sky for four years. It found that most stars in the Milky Way host at least one planet — meaning there are likely hundreds of billions of planetary systems in our galaxy alone.

9. Summary

Astrophysics is relentlessly humbling. Each answer reveals a deeper question. Dark matter, dark energy, the nature of the first stars, the origin of the highest-energy cosmic rays, the equation

Field	Key Equation	Central Idea
Cosmology	$v = H_0 d$	Universe expands; governed by Friedmann eqs.
Stellar Phys.	$L = 4\pi R^2 \sigma T^4$	Stars burn nuclei; live and die on the HR diagram
GR / Black Holes	$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$	Gravity is curved spacetime; BHs have horizons
Gravitational Waves	$h = \Delta L/L$	Ripples in spacetime from accelerating masses
High-Energy	$L_{\text{Edd}} \propto M$	Accretion and relativistic jets power AGN
Galactic Astro.	$v^2(r) = GM(r)/r$	Flat rotation curves $\Rightarrow$ dark matter halos
Exoplanets	$\delta = (R_p/R_\star)^2$	Transit photometry reveals distant worlds

Table 1: Key equations across astrophysical sub-fields.

of state of neutron stars — these remain open. The equations above are not the end of the story. They are the *vocabulary*. The sentences, the paragraphs, the story itself: that is still being written, night after night, by observers around the world and in orbit, and by theorists chasing the next beautiful equation.

“The universe is under no obligation to make sense to you.”

— Neil deGrasse Tyson